

Local SOP: CDU Overhead Corrosion Monitoring

SYNTHETIC DEMO DATA - NOT AN OPERATING PROCEDURE

Facility: Kavach Demonstration Refinery | Unit: Crude Distillation Unit (CDU)

1. Purpose

- Provide a demonstration knowledge source for reviewing corrosion observations in a crude distillation unit overhead system.

2. Roles

- Panel operator: monitor overhead temperature, pH, and chloride trend.
- Field operator: collect samples and verify chemical injection equipment.
- Inspection engineer: complete visual checks and ultrasonic thickness follow-up.

3. Operating controls

- Maintain wash-water injection at 5-7 percent of crude feed rate.
- When sour-water chloride exceeds 50 ppm, increase wash-water injection by 10 percent and record the adjustment.
- Maintain overhead accumulator boot-water pH between 5.5 and 6.5 through neutralizing-amine adjustment.
- Maintain filming-amine dosage according to the approved chemical programme.

4. Inspection response

- For localized pitting near a nozzle, schedule focused ultrasonic testing within 90 days.
- For chloride-attack indicators, review the chloride trend and wash-water rate with Operations before the next integrity review.
- Attach photographs and thickness readings to the integrity review pack.

5. Escalation

- If overhead pH falls below 4.0, notify the shift supervisor and apply the approved emergency response plan.
- This synthetic SOP does not authorize operational changes or replace site procedures.